

SUCHITRA SHANKAR

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EDUCATION

PES University (B.Tech, Computer Science & Engineering)

Bangalore, Karnataka

CGPA: 8.57, (CNR-Rao distinction for the top ~20%)

Aug 2023 – May 2027

PUBLICATIONS

- **Shankar, S.**, Rose, N. **Amethyst**: Workload-Aware Adaptive LSM Compaction. Published at FORMATS 2026 (co-located with **ACM SIGMOD** 2026). doi.org/10.1145/3802514.3812603

EXPERIENCE

Firebolt Analytics - R&D Intern

Jul 2026 – Oct 2026

- Developing automated evaluation frameworks for Firebolt and contemporaries, analyzing API, SDK, and tooling behavior across data workflows

Sansera Engineering Ltd - Intern

Jan 2026 – Apr 2026

- Developing XGBoost-based delay prediction models to optimize phase- and task-level scheduling in industrial pipelines, reducing latency variability and scheduling uncertainty, achieving ~**87%** accuracy.

The Innovation Lab - Research Intern

Jun 2025 – Aug 2025

- Designed and conducted an empirical study (MiCoZ) analyzing causal profiling behavior in asynchronous microservice DAGs. Conducted experiments on a distributed testbed in Go.

Twospoon.ai - Intern

Mar 2025 – Apr 2025

- Designed a novel intermediate representation for Java → Go transpilation and preserving semantic correctness

PROJECTS

Amethyst (github.com/amethystdb/amethyst_1.0)

Go • LSM trees

- Built adaptive **LSM-tree** with per-segment compaction policies driven by runtime access patterns (metadata + FSM controller).
- Achieved ~**2.2×** lower end-to-end runtime than tiered compaction while maintaining leveled-like read amplification and tiered-like write costs.

MiCoZ (theinnovationlab.in/projects?project=MiCoZ)

Go • Microservices

- Built Go-based microservice system for **causal profiling**; modeled async DAG execution under load, and evaluated latency propagation
- Measured **p95/p99** latency; used Mann–Whitney U tests to detect statistically significant bottlenecks.
- Focused on identifying non-obvious **bottlenecks** in distributed systems under realistic workloads. Paper under review at MASCOTS 2026.

MiniRaft (github.com/softserves/Mini-RAFT)

Go • Raft

- Implemented the **Raft consensus protocol** in Go using goroutines and channels for leader election, log replication, and replicated state synchronization across **distributed nodes**.
- Built RPC-based inter-node communication and synchronization mechanisms to **replicate** operations consistently while safely coordinating concurrent client requests.

TECHNICAL SKILLS

Languages: Go, C, Python, Java

Systems: Multithreading, Concurrency, Distributed systems, storage engines (LSM-trees), Parquet

Infrastructure & Networking: Git, Docker, Kafka, microservices, REST/gRPC, HTTP, TCP/IP

Performance & Evaluation: Benchmarking, profiling, p95/p99 latency analysis, statistical testing

ACHIEVEMENTS

Volunteer @ **Bengaluru Systems & Rootconf**

Ingenious Hackathon - Won **3rd place nationally** (100+ teams) for PawPatterns

Mentored Bento.rs (Rust container runtime) at Tilde 4.0; guided 30+ developers in **FOSS** contributions

Maintainer for NightManager and PhraseCraft during Hacknight (Hacktoberfest-style **FOSS** event)

Mentor at HackerSpace PESUECC, a student-run open-source developer community